GCSE · OCR · Computer Science

2 hours 32 questions

Identifying & Preventing Threats to Computer Systems & Networks

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Forms of Attack on a Network / Methods of Preventing a Network Attack

Easy (12 questions)	/15
Medium (11 questions)	/48
Hard (9 questions)	/42
Total Marks	**/105**

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Easy Questions

1 A software development company wants to **protect their computer systems** and **data** from unauthorised access.

Identify two methods of **physical security** that the company could use to protect their computer systems.

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..... **(2 marks)**

2 **State** the **definition** of the term '**social engineering**'.

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..... **(2 marks)**

3 **Identify the threat** penetration testing can be used to help prevent.

..... **(1 mark)**

4 **Define** the term 'Malware'

..... **(1 mark)**

5 **State** the primary goal of a **phishing** attack

..... **(1 mark)**

6 **Identify** the type of software targeted by an **SQL Injection**

..... **(1 mark)**

7 **State** the purpose of penetration testing

..... **(1 mark)**

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8 **Identify** two types of permissions managed by user access levels

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..... **(2 marks)**

9 **Define** the term 'Ciphertext

..... **(1 mark)**

10 **Identify** the device or software used by a hacker to collect data packets as they travel across a network.

..... **(1 mark)**

11 **Define** the method used in a **brute-force attack** to gain access to a system

..... **(1 mark)**

12 **State** the primary purpose of a **Denial of Service (DoS)** attack

..... **(1 mark)**

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Medium Questions

1 **Identify** and **describe** two **software-based security methods** that the company can use to protect their computer systems and data.

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..... **(6 marks)**

2 A hospital **stores patients' details on its computer network**.

The hospital is concerned about the security of its patients' details.

Staff **already use strong passwords** to protect systems.

Explain, with reference to system security, **three other ways** that the hospital could **protect the network** system.

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..... **(6 marks)**

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3 A hospital **stores patients' details on its computer network**.

The hospital is concerned about the security of its patients' details.

Staff **already use strong passwords** to protect systems.

Identify three errors that hospital staff could make that may **endanger the security of the network**.

Outline a procedure that could be put in place to **prevent each error**.

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..... **(6 marks)**

4 (a) A university want to protect their data against **threats** when connected to the Internet.

Describe the **threat malware can pose** to the university's network and **give a prevention method** that the university can use.

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..... **(3 marks)**

(b) **Describe** the threat a **brute force attack** can pose to the university's network and **give a prevention** method that the university can use.

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..... **(3 marks)**

5 (a) Data in computer systems is valuable and at **risk of loss**, **damage** or **being stolen**.

The table has **four potential threats** to data.

Write one prevention method for each threat below.

Each prevention method **must be different**.

- **Unauthorised access to a computer**
- **Virus**
- **Phishing**
- **Data interception**

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..... **(4 marks)**

(b) **Name two** other **threats** to the data in a computer system and **give a method** of **preventing** each.

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..... **(4 marks)**

6 Hamish stores **confidential documents** on his laptop.

Hamish needs his computer to be **secure from unauthorised access** when connected to a network.

Describe the **problems that can arise** from **unauthorised access to his laptop** and confidential documents.

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..... **(3 marks)**

7 **Describe two ways** Hamish can **help prevent unauthorised access** to his laptop.

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..... **(4 marks)**

8 Sarah works as a freelance graphic designer and stores **client projects** on her desktop computer.

Sarah's computer is **connected to the internet** and she is concerned about **cyber threats** that could compromise her work.

Describe the **security risks** that could affect **Sarah's computer and client projects** when connected to the internet.

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..... **(3 marks)**

9 **Describe** how a **brute-force attack** is carried out

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(2 marks)

10 **Explain** the impact of a **DoS attack** on a company's web server

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(2 marks)

11 **Describe** how a **firewall** monitors network traffic

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(2 marks)

Hard Questions

1 A house owner is concerned about **potential threats to the network** from being connected to the Internet.

Describe three possible threats to the computers connected to the network and **give one way** each threat can be **reduced** or **prevented**.

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..... **(9 marks)**

2 **Analyse** the potential consequences of a successful Distributed Denial of Service (**DDoS**) attack on a major e-commerce website and **explain** how this type of attack differs from a standard Denial of Service attack (**DoS**).

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..... **(5 marks)**

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3 **Evaluate** the **effectiveness** of penetration testing as a method for preventing network attacks. **Discuss** potential **limitations** and explain how organisations can maximise their benefits

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..... **(8 marks)**

4 **Compare** the spreading mechanisms of a computer **virus** and a **worm**

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..... **(4 marks)**

5 **Evaluate** why the "human element" is often the weakest link in network security

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..... **(4 marks)**

6 **Justify** the use of **input validation** as a primary defence against SQL Injection

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..... **(3 marks)**

7 **Explain** why anti-malware software must be **regularly updated**

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..... **(2 marks)**

8 **Evaluate** the use of **biometrics** over traditional keys for a server room

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..... **(4 marks)**

9 **Analyse** why data transmitted over a wireless connection is more vulnerable to interception than data on a wired connection

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..... **(3 marks)**

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